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BLOOMBERG SCHOOL
of PUBLIC HEALTH

Numeric Estimates in Epidemiology

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Learning Objectives

- ▶ Describe numerators in public health data
- ▶ Explain denominators in public health data, and where this data can be found
- ▶ Calculate ratios and proportions to describe public health problems
- ▶ Use rates to describe the burden of an issue in a population



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Numerators

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What Is a Numerator?

- ▶ Counts
 - ▶ Events
 - ▶ Score
 - ▶ People
- ▶ As a mathematical concept: the top number in a fraction

Numerator
Denominator

▶ Example: $\frac{1}{2}$ or $\frac{3}{4}$

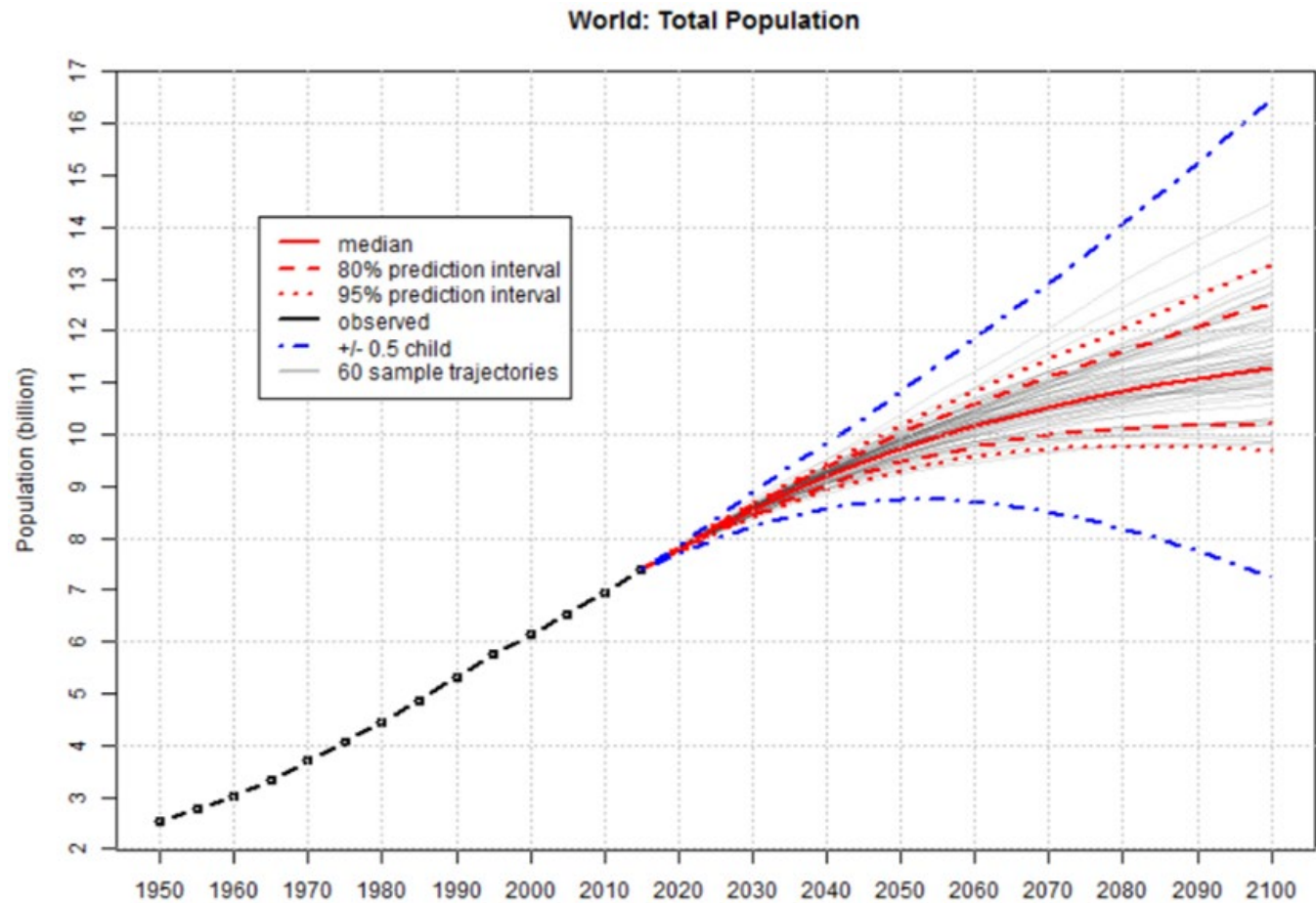
Numerators in Public Health

- ▶ Definition: The ***count*** of persons with a certain trait or condition or who use a certain program or intervention
- ▶ Examples:
 - ▶ How many people are served by a particular substance use intervention program?
 - ▶ How many children suffer from symptoms of post-traumatic stress disorder following exposure to a natural disaster?

Why Do We Care?

- ▶ We can measure how big of a problem a particular disease is in our community
- ▶ We can compare our numbers with another population or area
- ▶ We can decide how to allocate our resources
- ▶ We can see changes over time

Example: United Nations World Population Projections

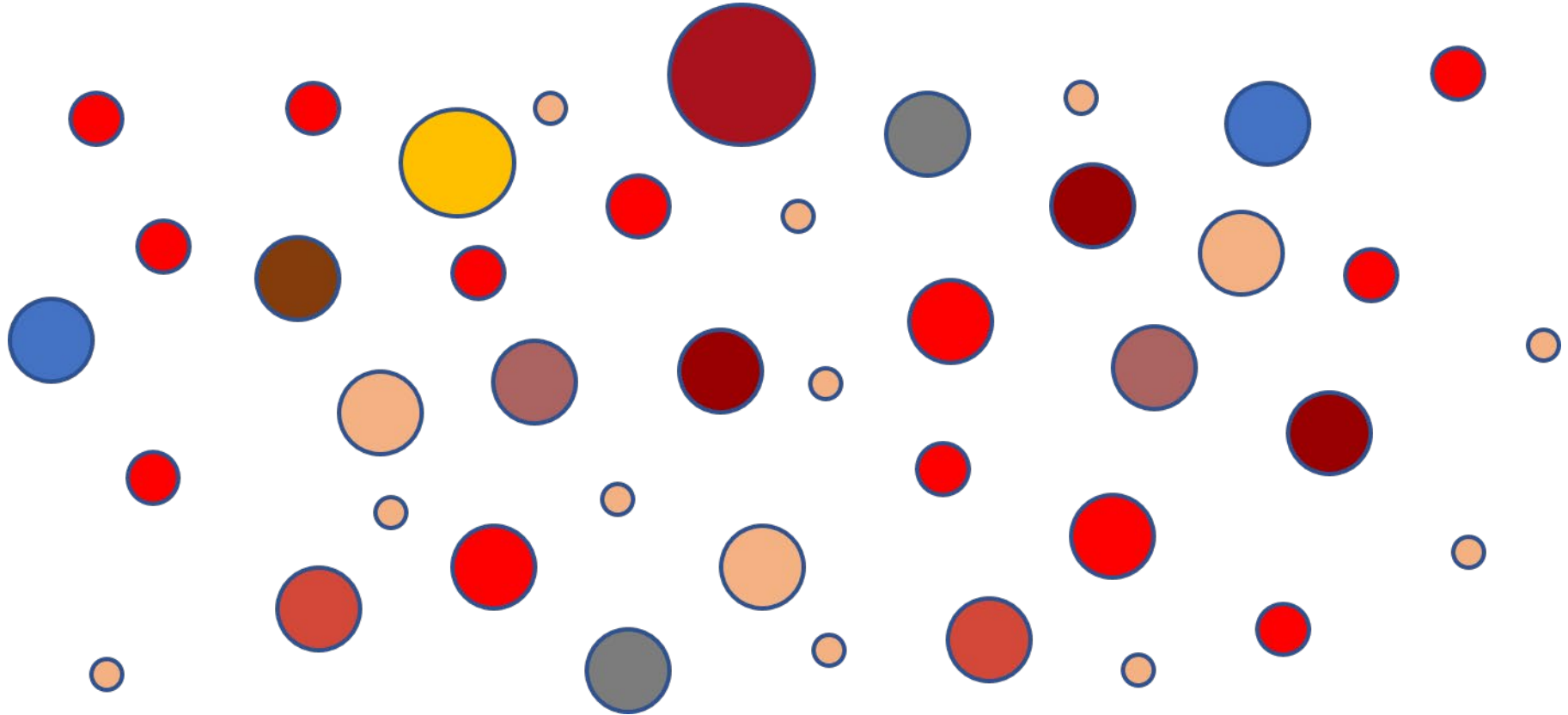


Source: United Nations, Department of Economic and Social Affairs, Population Division.
(2017). *World population prospects: The 2017 revision*. Retrieved October 12, 2018.

Where Can We Get Numerators?

- ▶ From surveys or registries
 - ▶ Census
 - ▶ Telephone book
 - ▶ Disease registry
- ▶ Our own counts
 - ▶ Someone's observations
 - ▶ A formal study

Exercise: How Many Small Dots?



Case Definition

- ▶ Clearly define who or what gets counted
 - ▶ What/who is included
 - ▶ What/who is excluded
- ▶ Examples (most commonly from outbreaks)
 - ▶ Watery diarrhea: more than three watery non-bloody stools per day
 - ▶ Lower respiratory infection: fever, cough, and rapid breathing (>50 breaths/minute)

Utility of Case Definitions

- ▶ Helps you decide who to count in your numerator
- ▶ Allows you to differentiate people with “similar” but not exact symptoms
- ▶ Example: number of students enrolled in this course
 - ▶ Exclude someone who is working on something else while this lecture plays in the background
 - ▶ Exclude people who do not complete the full lecture series



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Denominators

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What Is a Denominator?

- ▶ As a mathematical concept: the bottom number in a fraction
- ▶ The total population pool
 - ▶ People, jellybeans, hospital admissions, trees, salary (in dollars), car crashes, etc.

Numerator

Denominator

- ▶ Example: $\frac{1}{4}$ or $\frac{29}{37}$
- ▶ Etc.

Denominators in Public Health

- ▶ Numerator = health event measure
- ▶ **Denominator** = population from which events are drawn
- ▶ Examples:
 - ▶ If measuring incidence of pregnancy among teens, who might comprise your denominator?
 - Young women 13–19 years of age in your population

Why Do We 'Care' about Denominators in Public Health?

- ▶ The reference population from which health events are drawn and measured
- ▶ The total population in an area over a defined time
- ▶ Used for reference and comparison to give an idea of scale and scope

Where does Denominator data come from?

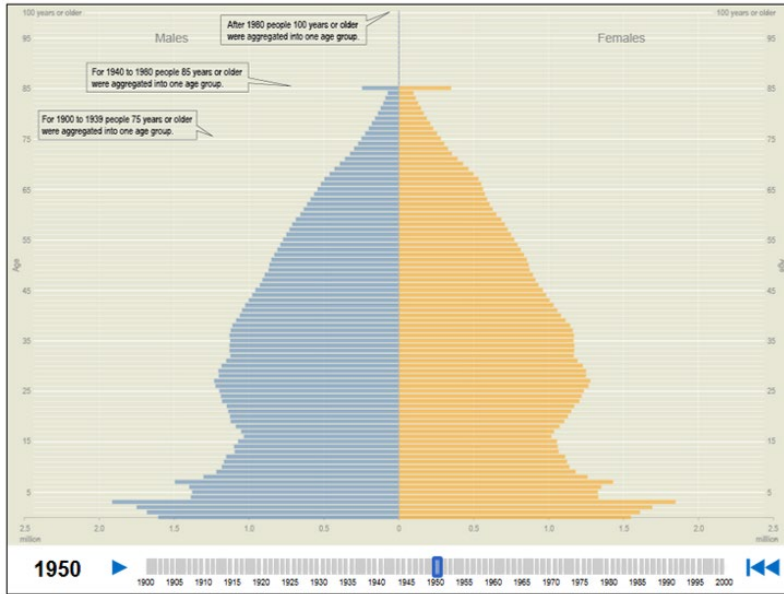
- ▶ **Collect** the data (yourself)
- ▶ **Obtain** the data (from somewhere else)
 - ▶ Census
 - ▶ Local school or ID registry
- ▶ International population statistics:
 - ▶ World Health Organization
 - ▶ United Nations World Factbook
 - ▶ Research agencies (i.e., Demographic and Health Surveys Program)

Know Your Population

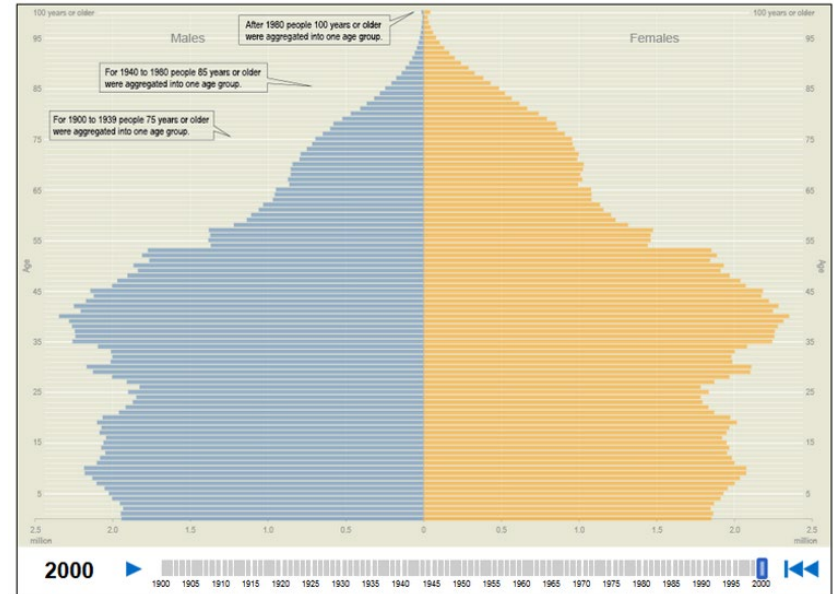
- ▶ Having a sense of who is in the catchment area in which you are interested is key
- ▶ Knowledge of the population structure can be important to policy-making and planning
- ▶ It can also help inform the allocation of resources

Compare and Contrast: US Age/Sex Population Structure over Time

1950 US population: 152.3 million



2000 US population: 282.2 million



Source: US Census Bureau. (2013, September 12). A century of population change in the age and sex composition of the nation. Retrieved October 12, 2018.

What Can Be Counted?

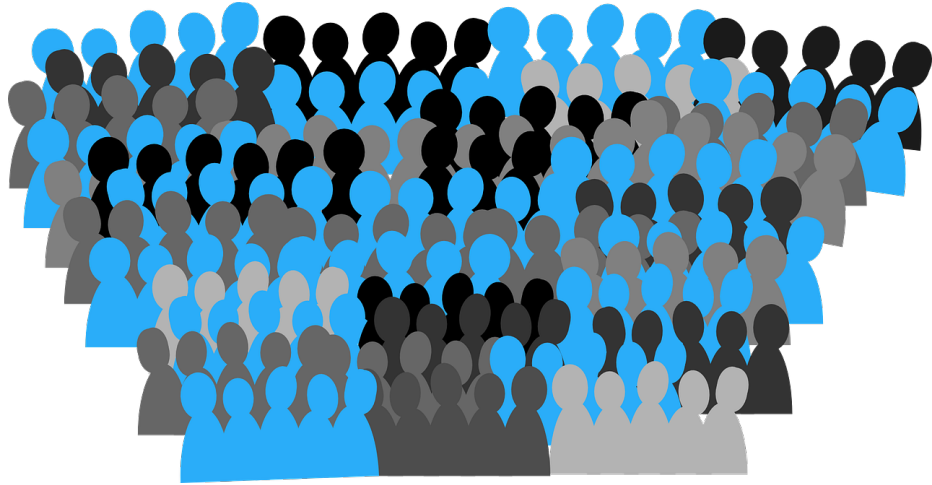
- ▶ People, communities, entire populations, etc.
- ▶ What you use as your denominator relates back to what you are measuring or planning
- ▶ Note 1: In order to be counted in the numerator as a measured event, an individual must be in the denominator
- ▶ Note 2: You can only measure (events) that can occur in your denominator

Choices of Denominators Might Change

- ▶ There's a new test to identify health condition K
 - ▶ You are tasked with deciding how many test kits to purchase
- ▶ Who is your denominator?
 - ▶ Number of people in your region?
 - ▶ Number in the age/sex subgroup at highest risk?
- ▶ You learn that although everyone is eligible to access the test, only one health insurer for the region covers the test
- ▶ Now who is your denominator?
 - ▶ Number covered by that health insurance company?

Identifying a Denominator Can Be Difficult

- ***Remember:*** not everybody wants to stand up and be counted



Exercise: Identify the Numerator and the Denominator—1

- ▶ Community X has a total population of 100,000 people
 - ▶ 600,000 males (60,000 are 13–19 years of age)
 - ▶ 400,000 females (40,000 are 13–19 years of age)
- ▶ You are looking at episodes of teen pregnancy in your population
- ▶ What is your numerator?
- ▶ What is your denominator?

Exercise: Identify the Numerator and the Denominator—2

- ▶ Community X has a total population of 100,000 people
 - ▶ 600,000 males (60,000 are 13–19 years of age)
 - ▶ 400,000 females (40,000 are 13–19 years of age)
- ▶ You are looking at episodes of teen pregnancy in your population
- ▶ What is your numerator?
 - ▶ **Number of pregnant females aged 13–19 years**
- ▶ What is your denominator?
 - ▶ **Total number of females aged 13–19 years (40,000)**



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Ratios and Proportions

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Ratio

- ▶ Used to compare two (or more) numbers
- ▶ The ratio can be written as:
 - ▶ A:B
 - ▶ A/B
- ▶ Both terms can be multiplied or divided by the same number and still represent the same ratio
 - ▶ Example: Community X has 600,000 males and 400,000 females
 - ▶ The ratio of males to females can be expressed as:
 - 600,000:400,000
 - 600:400
 - 3:2

Proportion—1

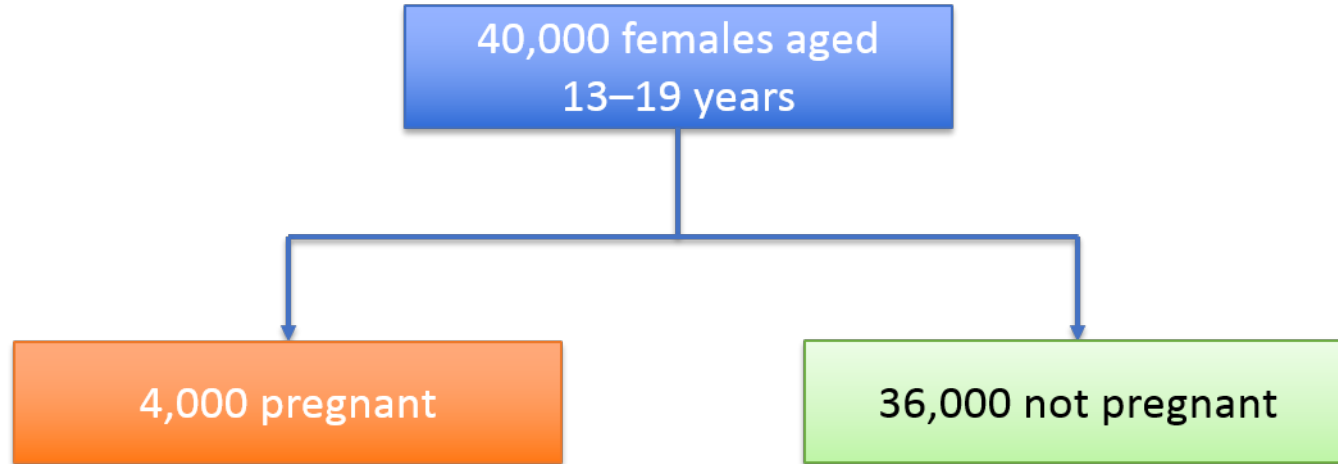
- ▶ A proportion relates a part to a whole
- ▶ In a proportion, the numerator is a part of the denominator
- ▶ Example: Community X has 600,000 males and 400,000 females
 - ▶ The proportion of males in Community X is $600,000/1,000,000$
 - ▶ The proportion of females in Community X is $400,000/1,000,000$
- ▶ For ease of communication and comprehension, proportions are often expressed as “per 100” or “percent”
 - ▶ So, the proportion of males is $600,000/1,000,000$ or $60/100$ or 60%
 - ▶ What is the proportion of females expressed as a percent?

Proportion—2

- ▶ A proportion relates a part to a whole
- ▶ In a proportion, the numerator is a part of the denominator
- ▶ Example: Community X has 600,000 males and 400,000 females
 - ▶ The proportion of males in Community X is $600,000/1,000,000$
 - ▶ The proportion of females in Community X is $400,000/1,000,000$
- ▶ For ease of communication and comprehension, proportions are often expressed as “per 100” or “percent”
 - ▶ So, the proportion of males is $600,000/1,000,000$ or $60/100$ or 60%
 - ▶ **The proportion of females is $400,000/1,000,000$ or $40/100$ or 40%**

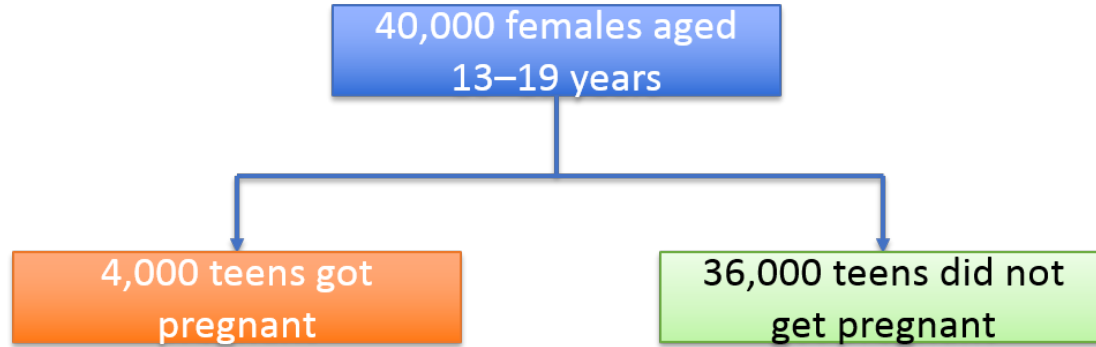
Exercises—1

Community X (2010–2015):



Exercises—2

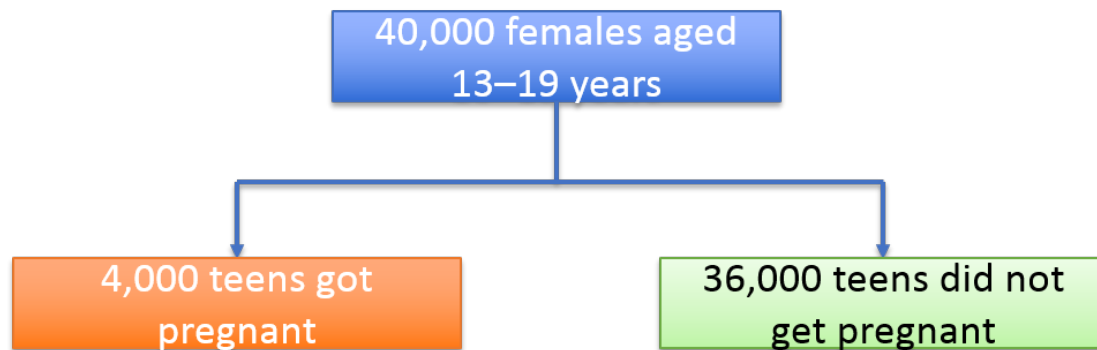
Community X (2010–2015):



- ▶ What was the ratio pregnant to non-pregnant teens in 2010–2015?
- ▶ What proportion of female teens were pregnant in 2010–2015?

Exercises—3

Community X (2010–2015):



- ▶ What was the ratio pregnant to non-pregnant teens in 2010–2015? **4,000:36,000 or 1:9**
- ▶ What proportion of female teens were pregnant in 2010–2015? **4,000/40,000 or 10%**



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Rates

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Rates: Overview

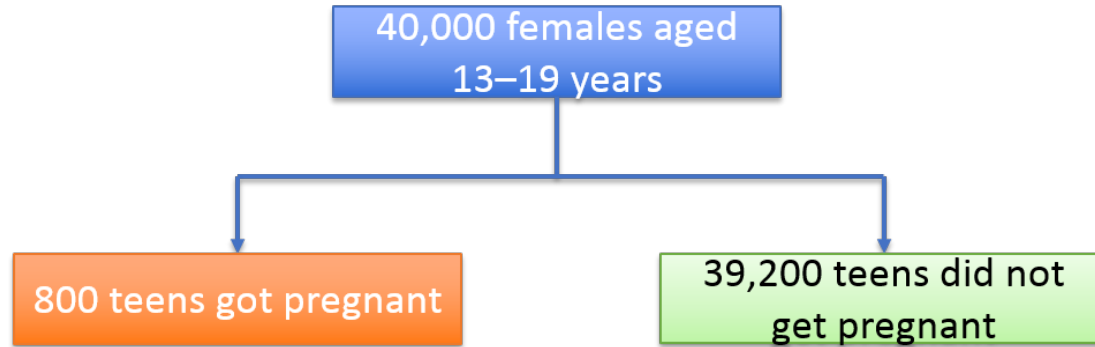
- ▶ Ratio of:

$$\frac{\textit{Numerator}}{\textit{Denominator}}$$

- ▶ **Includes time**, which is the same in both
- ▶ In public health, the unit of time is often one year
 - ▶ Many times, we get sloppy in expressing rates, and do not explicitly say “per year”
 - ▶ Example: if there were 20 deaths in a population of 1,000 in 2018, the death rate is often expressed as 2%, but in actuality, it is 2% *per year* OR *in 2018*

Exercises—1

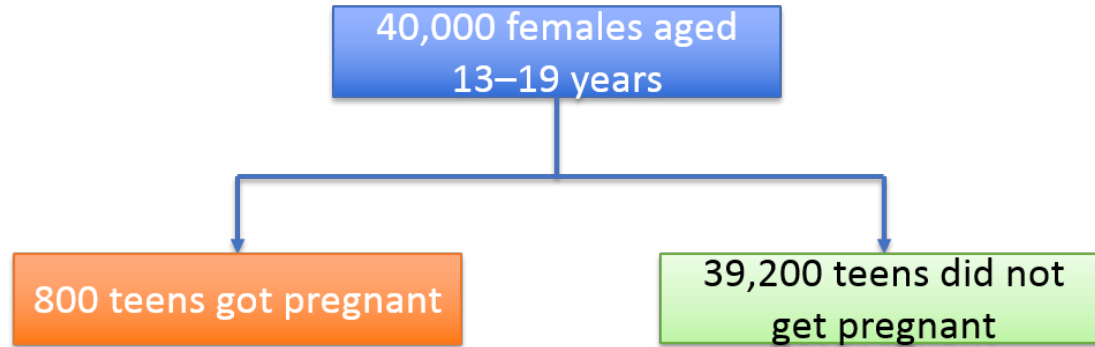
Community X (**2012**):



- What is the rate of teen pregnancy in 2012?

Exercises—2

Community X (**2012**):



- What is the rate of teen pregnancy in 2012? **$800/40,000$ or 0.02 in 2012**

Standard Measure of Rates—1

- ▶ Usually expressed as “per certain population size”
 - ▶ Example: percent (per 100 individuals)
 - ▶ Example: per 100,000 individuals
- ▶ Allows a standard population measure to be easier to express and to allow comparison
 - ▶ Example: teen pregnancy rate in Community X in 2012
 - ▶ $800/40,000$ or **0.02 in 2012**
 - ▶ $0.02 \times 100 = \mathbf{2\% \text{ of teens in 2012}}$
 - ▶ $0.02 \times 1,000 = \mathbf{20 \text{ per 1,000 population of teens in 2012}}$
- ▶ Exercise: How would you express each of these rates in a phrase?

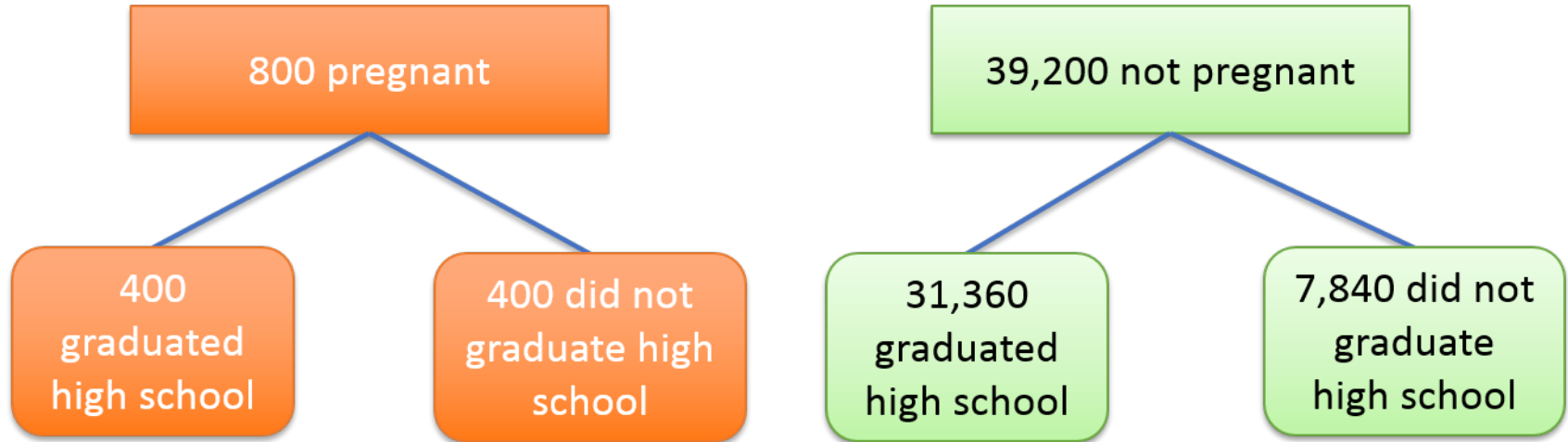
Standard Measure of Rates—2

- ▶ Usually expressed as “per certain population size”
 - ▶ Example: percent (per 100 individuals)
 - ▶ Example: per 100,000 individuals
- ▶ Allows a standard population measure to be easier to express and to allow comparison
 - ▶ Example: teen pregnancy rate in Community X in 2012
 - ▶ $800/40,000$ or 0.02 in 2012 **0.02 pregnancies per teen female in 2012**
 - ▶ $0.02 \times 100 = 2\%$ in 2012 **2% of teen females had a pregnancy in 2012**
 - ▶ $0.02 \times 1,000 = 20$ per 1,000 in 2012 **20 pregnancies per 1,000 teen females in 2012**

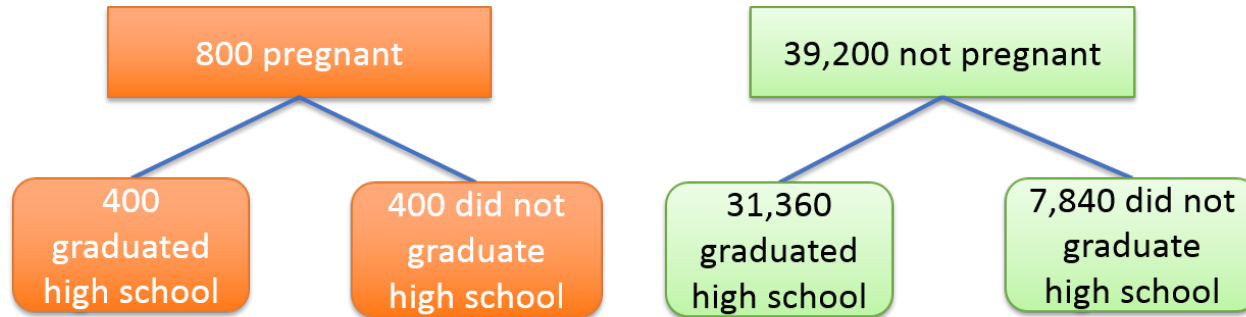
Using Rates for Comparison

- ▶ Rates allow standardization of a measure so that two populations (or two scenarios) can be compared
- ▶ Question: In Community X, is teen pregnancy associated with a decreased chance of obtaining a high school diploma?
 - ▶ To answer this: we looked at our group in 2012 and followed them forward through 2018 to see if they obtained their high school diploma

Exercise: Using Rates for Comparison—1

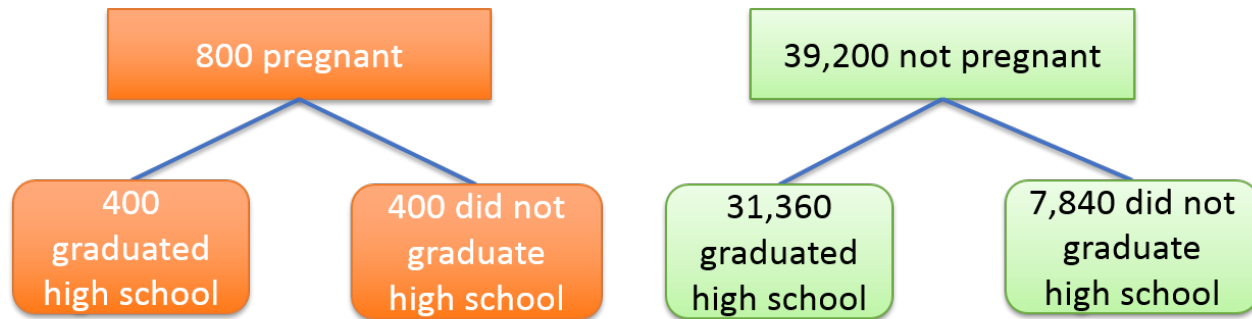


Exercise: Using Rates for Comparison—2



- ▶ Question: is teen pregnancy associated with a decreased chance of obtaining a high school diploma?
- ▶ Calculate the high school graduation rates among teens who got pregnant and those who did not
 - ▶ Remember to standardize the population; for this example, use “per 1,000”

Exercise: Using Rates for Comparison—3



- ▶ Graduation rate, teens who got pregnant: $400/800 = 500$ per 1,000 by 2018
- ▶ Graduate rate, teens who did not get pregnant: $31,360/39,200 = 800$ per 1,000 by 2018

Alternate Denominator for Calculating Rates

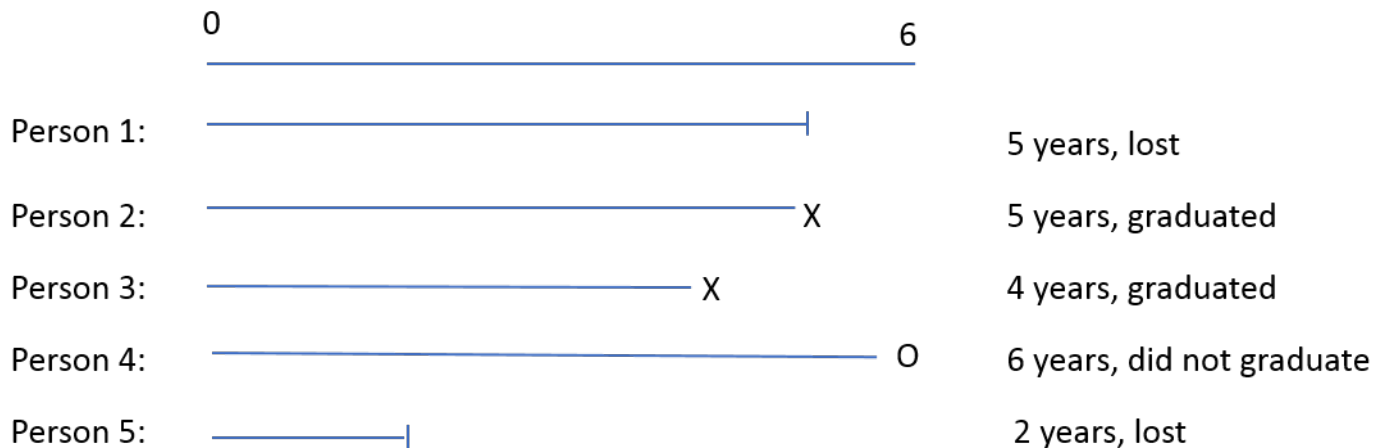
- ▶ So far, we have discussed calculation of rates using “number of people in a population” as a denominator
- ▶ However, in a study setting where you are following a group of individuals over time, an alternative option is to calculate the “amount of observation time your group (or cohort) contributed to your study”
 - ▶ Consider our example of observing teens from 2012 to 2018 to see if they obtained their high school diploma:
 - Some of those teens might have left our community
 - Some of those teens might have fallen off of our radar (so we don’t know what happened to them)

Denominator for an Observed Cohort

- ▶ The denominator can be expressed as the “person-time” at risk of an event
 - ▶ Allows for inclusion of observation time of those who left your study or had an alternate outcome, such as death
 - ▶ Results in an accurate calculation of the rate of an event in a particular population “at risk” for that event
- ▶ Most common is person-years, but you can also do person-days, person-months, or any other measure
- ▶ The person-time denominator is the amount of time that each person is observed during which they are able to achieve the outcome you are observing

Example: High School Graduation Rates among Pregnant Teens—1

Observation period: six years

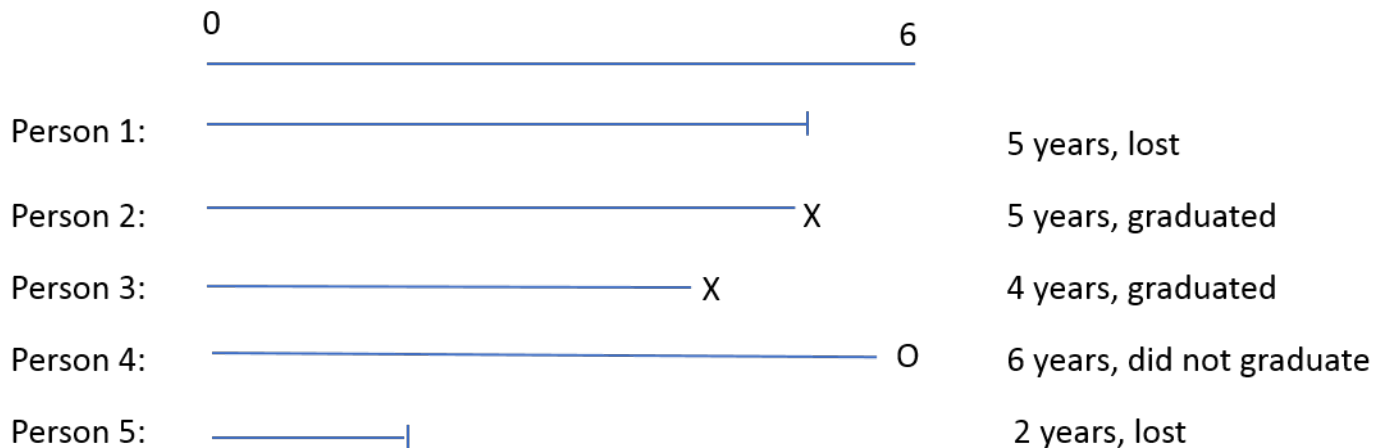


Example: Calculation of Rate per Person-Time—1

- ▶ Time period of six years; event = high school graduation
 - ▶ Two people had the event of interest
- ▶ High school graduation rate = $\# \text{ graduated} / (\text{total person-time})$
- ▶ Calculation of person time
 - ▶ = $5+5+4+6+2$ person-years = 22 person-years
- ▶ How many people graduated?

Example: High School Graduation Rates among Pregnant Teens—2

Observation period: six years



Example: Calculation of Rate per Person-Time—2

- ▶ Time period of six years; event = high school graduation
 - ▶ Two people had the event of interest
- ▶ High school graduation rate = # graduated/(total person-time)
- ▶ Calculation of person time
 - ▶ = 5+5+4+6+2 person-years = 22 person-years
- ▶ High school graduation rate: **2/22 person-years = 0.09 per person-year or 9 per 100 person-years**

What Can We Do with Rates?

- ▶ Compare a rate measured today with...
 - ▶ The rate at another time
 - ▶ The rate in a different population
- ▶ Compare findings from other studies
- ▶ Observe trends over time

Summary

- ▶ Numerators are counts of events or people
- ▶ Denominators comprise the population from which numerator events are drawn
- ▶ Ratios compare two or more numbers
- ▶ Proportions relate a part to a whole
- ▶ Rates are ratios with time implicit in the denominator

Thank you!